# Math 1232 Summer 2025
# Mastery Quiz 3
# Due Thursday, July 10

This mastery quiz has three topics. Everyone should do M2, but if you already have a 4/4 on M1 or a 2/2 on S2, feel free to skip these. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write "yes" or "no" or give a single number.

Please turn in this quiz in class on Thursday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

**Topics on this quiz:**

- Major Topic 1: Transcendental Functions
- Major Topic 2: Integration Techniques
- Secondary Topic 2: L'Hôpital's Rule

# Name:

Name: ________________________________________

## Major Topic 1: Transcendental Functions

(a) Compute $\displaystyle\int 3^x(4+3^x)^3\,dx$.

(b) Compute $\dfrac{d}{dx}\left(\sqrt{x+1}\right)^x$.

(c) Compute $\dfrac{d}{dx}\arccos(x+e^x)$.

Name:

## Major Topic 2: Integration Techniques

(a) Compute $\displaystyle\int \sin(2x)\cos(3x)\,dx$.

(b) Compute $\displaystyle\int \cos^5(2x)\,dx$.

Name: ________________________________

(c) Compute $\displaystyle\int \frac{dx}{x^2\sqrt{4-x^2}}$.

Name:

## Secondary Topic 2: L'Hôpital's Rule

(a) Compute $\lim_{x \to 0} \left( \frac{e^x + 1}{2} \right)^{1/x}$.

(b) Compute $\lim_{x \to +\infty} \frac{\arctan(x)}{\arctan(x) + 1}$.

(c) Compute $\lim_{x \to 1} \frac{\ln(x)}{\arcsin(2x - 2)}$.